

Probability Distribution:-

- ① Random Experiment - whenever we perform an experiment under very nearly identical condition, we expect to obtain results which are essentially the same. There are results which are not same where condⁿs may be nearly identical.
for ex: we throw a die.

Outcome maybe any one from the same $= \{1, 2, 3, 4, 5, 6\}$.

Each time, we perform this experiment, the result may be uncertain. Such a experiment is called random experiment.

- ② Outcome - Result of the random

E.g $\rightarrow$ Throw Die $\rightarrow$ Six possible outcome from the experiment.

$$\{x \mid 1 \leq x \leq 6, x \in \mathbb{Z}\}$$

- ③ Event $\rightarrow$ Any phenomenon that occurs in an random Experiment is Event.
An Event can be composite or elementary.

Elementary $\rightarrow$ Single Outcome.

Composite $\rightarrow$ More than single outcome.

$\hookrightarrow$ Ex: Throw Die.

- ④ Impossible Event $\rightarrow$ A Event which cannot occur in a random Experiment.

Ex: 7 in Die.

(iv) Certain Event: Event which is sure to occur called certain event

(v) Complementary Event: Two events are complement s.t., one of the event is negation of other.

Ex: {Getting a six}

Ex: {Not ————}

(vi) Mutually Exclusive: Events where happening of one event doesn't include happening of all others.

Elementary Events: {1, 2, 3, ————}

(vii) Equally Likely Event: Two events are called equally likely if one of events cannot happen in preference of other.

Ex: Tossing Coin.

→ "Occurring head" & "Occurring tail"

(ix) Exhaustive Event: A set of events are said to be exhaustive if all possible outcome.

One of which necessarily occurs.

Ex: Tossing coin → "Occurring head" & "Occurring tail"

(vi) Random Experiment: Set of all possible outcomes in a random experiment.

(xii) $P \in C$.

(xiii) Random Experiment: for a random experiment, "n" mutually exclusive & "n" equally likely cases. ~~there~~ are favourable to the happening of an event E ("say"), then probability (P) of happening E ,

$$P = \frac{n}{N} \quad \text{i.e.,} \quad \frac{\text{no. of favourable space}}{\text{Sample space}}$$

$$[0 \leq P \leq 1] \text{ or, } P \in [0, 1].$$

find Probability of getting multiple of 2 when a die is thrown up.

$$\text{Sample} = \{1, 2, 3, 4, 5, 6\}$$

$$\text{favourable} = \{2, 4, 6\}$$

$$P = \frac{3}{6}$$

(xiv) Axiomatic Probability: for any event E ,

(i) Probability of $\frac{P}{E} \geq 0$

(ii) If S is sample space of experiments, i.e. set of all possible outcome.

$$\text{then } P(S) = 1$$

(iii) If X & Y are mutually exclusive outcomes $P(X \cup Y) = P(X) + P(Y)$

$$X \in S,$$

$$Y \in S.$$

Thus, Axiomatic Defn: $0 \leq P(E_i) \leq 1$

$$E_i \in S, \forall i$$

$$(ii) \sum_i P(E_i) = 1$$

$$\emptyset \subset S.$$

$$(iii) P(\emptyset) = \sum_{E_j \in \emptyset} P(E_j)$$

$$E_j \in \emptyset.$$

* Addition Theorem:-

If 'n' events $E_1, E_2, \dots, E_n$ be the probability $P(E_1), P(E_2), \dots, P(E_n)$

for any two events E_i & E_j

$$E_i \cap E_j$$

$$P(E_i \cup E_j) = P(E_i) \cup P(E_j)$$

$$P(E_i) + P(E_j) - P(E_i \cap E_j)$$

for 'n' no. of events, probability of any event occurs:-

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i) + P(E_j) - P(E_i \cap E_j)$$

$$+ \sum_{\substack{i=1 \\ j=1 \\ k=1 \\ i \neq j \neq k}} P(E_i \cap E_j \cap E_k)$$

$$1 \neq j \neq k$$

$E_1, E_2, \dots, E_n \rightarrow$ Mutually Exhaustive,

Atleast two of them simultaneously occur.

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$$

$$\text{i.e., } P(E_1 \cup E_2 \cup \dots \cup E_n)$$

$$= P(E_1) + P(E_2) + \dots + P(E_n)$$

Compound Probability: If two events E_1, E_2 happen simultaneously, then they are compound events.

Eg: Two dice are thrown simultaneously then can give any random number.

If two events occur simultaneously,

Eg: A person draws a ~~card~~ card, any event is not dependent on the first.

* Dependent Events: The events that are not mutually exclusive.

Ex: Person draws a ball from an urn & doesn't replace, any drawn by one person is not i.e., if E_1 & E_2 are dependent events then E_1/E_2 (denotes the prob. of occurrence of E_2)

If E_1 and E_2 are corresponding events, then the theorem of Conditional Probability,

$$P(E_1/E_2) = P(E_1) \times P(E_2),$$

if E_1 & $E_2 \rightarrow$ independent

$$P(E_1) \times P(E_1/E_2) \rightarrow \text{if } E_1, E_2 \rightarrow \text{dependent}$$

Thus, E_1 and E_2 as dependent events

$$P(E_2|E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} \quad [P(E_1) \neq 0]$$

$$\text{Again, } P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} \quad [P(E_2) \neq 0]$$

$$\therefore, P(E_1 \cap E_2) = P(E_2|E_1) \times P(E_1) \\ = P(E_1|E_2) \times P(E_2)$$

Thus, $P(E_1|E_2)$ or $P(E_2|E_1)$ are called conditional prob. of E_1 (or E_2) if E_2 (or E_1) has already occurred.

Bayes' Theorem

Let $E_1, E_2, E_3, \dots, E_n$ are mutually exclusive events associated with sample space S , where $P(E_i) \neq 0$, and $E_1, E_2, \dots, E_n$ form a partition of S .

Now, A is any event from S with $P(A) \neq 0$.

$A \cap E_i$ is also mutually exclusive for all i .

~~So, $A = A \cap \bigcup_{i=1}^n E_i$~~

$$A = A \cap \bigcup_{i=1}^n E_i = \bigcup_{i=1}^n (A \cap E_i)$$

$$P(A) = P\left(\bigcup_{i=1}^n (A \cap E_i)\right)$$

[By distributed law]

$$= \sum_{i=1}^n P(A \cap E_i) = \sum_{i=1}^n P(E_i) \times P(A|E_i)$$

$$\text{Now, } P(E_i|A) = \frac{P(A \cap E_i)}{P(A)} = \frac{P(A \cap E_i)}{\sum_{i=1}^n P(A \cap E_i) \times P(A|E_i)}$$

Q1. A bag contains ⁶W & ⁴B. Another bag 2 contains 4W & 3B. 1 ball is drawn at random & it is found to be black. Find prob. that it is drawn from bag 1.

Soln: Let E_1 = Event of choosing bag 1
 E_2 = " " " " " bag 2.

$$\text{So, } P(E_1) = P(E_2) = \frac{1}{2}$$

$$P(A|E_1) = P(\text{drawn B from bag 1}) = \frac{4}{10}$$

$$P(A|E_2) = P(\text{drawn B from bag 2}) = \frac{3}{7}$$

$$P(E_1/A) = \frac{P(A|E_1) \times P(E_1)}{P(A|E_1) \times P(E_1) + P(A|E_2) \times P(E_2)}$$

Q2. In repeated trials, prob. of ~~success~~ success in ${}^nC_r p^r q^{n-r}$ where n = no. of trials & p is prob. of ~~success~~ succ. & q is prob. of failure. Thus, prob. of a success at n trials

$$= \sum \text{of prob. of a success}$$

$$= \sum {}^nC_r p^r q^{n-r} + {}^nC_{r+1} p^{r+1} q^{n-r-1} \dots {}^nC_n p^n$$

P1 An urn has 3 W & 5 B. 1 ball is drawn. Find the prob. that it is black.

Soln: Reqd. prob. = $\frac{1 \text{ black in 5 ball}}{1 \text{ ball in 8 ball}} = \frac{5C_1}{8C_1}$

P2 Two dice are thrown simul. Find prob. to get a total score of 7.

Soln: For throwing 2 dice, sample has 36 spaces.
Possible: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)
[6 ways]
Reqd. prob. = $\frac{6}{36} = \frac{1}{6}$

P3 6 boys & 6 girls sit in a row randomly. Find P:
i) 6 girls sit together
ii) boys & girls sit alternately.

Ans: i) 6 boys & 6 girls can sit randomly = $12!$ ways
Considering, 6 girls together as boys
as object = $7!$ [where 6 girls can arrange in $6!$ ways]
 $\therefore$ Reqd. prob. = $\frac{7! \times 6!}{12!} = \frac{1}{132}$

ii) For alternate boys & girls, can be arranged in $6! \times 6!$ ways [Girls together as an object]
1 boy & 1 girl can arrange in $2!$ ways
 $\therefore$ Prob. = $\frac{6! \times 6! \times 2!}{12!} = \frac{1}{462}$

P4 If 2 dice are thrown, Find P s.t. it show > 3.

Soln: ~~All dice is state P for dice~~
Let A be prob. that first dice show > 3.
Let B " " " " that second dice show > 3.

$$P(A) = \frac{3}{6}, P(B) = \frac{3}{6}$$

$$\therefore \text{Total} = \frac{9}{36} = \frac{1}{4} \quad \text{Total 9, check } (A \cap B)$$

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

P8 An article manufactured by a company consists of 2 parts x and y. In the process of manufacturing of part x, 9/10 defective. Sim., 5/100 are defect in part y. Find that defective.

Soln: $P(X_{\text{def}}) = \frac{9}{104}$, $P(\bar{X}_{\text{def}}) = 1 - \frac{9}{104} = \frac{95}{104}$

$$P(Y_{\text{def}}) = \frac{5}{100}, P(\bar{Y}_{\text{def}}) = 1 - \frac{5}{100} = \frac{95}{100}$$

$$\therefore P(\bar{X}_{\text{not def}}) \times P(\bar{Y}_{\text{not def}}) = \frac{95}{100 \times 104} = \frac{368}{416}$$

P6 A prob. in maths is given to 3 student, chance $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. Prob. that problem is solved.

Soln: $P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{3}, E_3 = \frac{1}{4}$, where E_i is event of selecting student 1, 2, 3.

$$\begin{aligned}\therefore P(\text{problem will be solved}) &= P(\text{none of them will succeed}) \\ &= 1 - P(\bar{E}_1) \times P(\bar{E}_2) \times P(\bar{E}_3) \\ &= 1 - 1\end{aligned}$$

P7 3 faces of a die are yellow, 2 faces ^{red & 1 blue} → blue. Dice → 3 times, find prob. that color yellow, red & blue appear respec.

Soln Let E_1 = Color yellow appear 1st
 E_2 = Color red appear 2nd
 E_3 = Color blue appear 3rd
 $P(E_1) = \frac{3}{6}, P(E_2) = \frac{2}{6}, P(E_3) = \frac{1}{6}$

∴ All outcome indepen.

$$P(E_1 \times E_2 \times E_3) = \frac{3}{6} \times \frac{2}{6} \times \frac{1}{6} = \frac{1}{36}$$

P8 The contents of arm I, II & III are as follows:
 1 W, 2 B, 3 R; 2 W, 1 B, 1 R; 4 W, 5 B, 3 R balls. 1 arm is chosen at random & 2 balls are drawn. They are W & R. Find P that it's II.

Soln A & 2 balls drawn from selected arm are white & red.

E_1 : Arm I, E_2 : II, E_3 : III

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

$$\begin{aligned}\text{Also, } P(A|E_1) &= \text{Prob. of getting W \& R when I is chosen} \\ &= \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2}\end{aligned}$$

$$P(A|E_2) = \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2}$$

$$P(A|E_3) = \frac{{}^4C_1 \times {}^3C_1}{{}^{12}C_2}$$

$$P(E_2|A) = \frac{P(E_2) \times P(A|E_2)}{\sum_{i=1}^3 P(E_i) \times P(A|E_i)} = \frac{\frac{1}{3} \times \frac{1}{3}}{\frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{11}}$$

$$\sum_{i=1}^3 P(E_i) \times P(A|E_i)$$

$$= \frac{58}{118}$$

Prob. questions

- Q1. There are 3 events A, B, C. only 1 can happen, the odds are 8 to 3 against A, 5 to 2 - B. Find odds against C.
 [Ans: 43/44]
- Q2. Given, P: A can solve is $\frac{2}{3}$, B can solve is $\frac{3}{5}$. Find P:
 i) Atleast A or B can solve the problem [13/15]
 ii) None of the 2 will solve [2/15]
- Q3. There are 2 groups, ~~one~~ is Science & 3 eng, & other has 3 science and 5 eng. An unbiased die is cast if no. 3 or no. 5 turns up a sub. is selected at random from 1st otherwise 2nd group. Find P that an eng. is selected from 1st group. [13/24]
- Q4. In 2000, there will be 3 candidates for position of principal, Mr. X, Y, Z whose chance of getting appoint. are in proportion 4:2:3. Prob. that Mr. X if selected will introduce co-education is 0.3. The prob. of Mr. Y & Mr. Z win the same are 0.5 & 0.8. Find P that three will bring co-education. [43/45]
- Q5. In a bolt factory, ~~ABC~~ Machine 1, 2, 3 manufacture 25%, 35%, 40% of the total. Of their output, 5, 4, 2% are defective bolts. A bolt is drawn at random & found to be defective. P that it is manufactured by machine 1, 2 & 3.
- Q6. In a test, an examinee either guesses or copies the answer of 4 mcq with 4 choice. Prob. that he guesses is $\frac{1}{3}$ & P that he copies is $\frac{1}{6}$ & P that his answer is correct & copied is $\frac{1}{8}$. Find P that he knew the answer given he correctly answered. [24/29]

Q7. A bag has 1 B & 2 W balls. A drawing from the bag consists of taking a ball from bag & keeping it out of it is which but putting it back is black. Find P:

- i) 1st drawing is W
- ii) 2nd " " W
- iii) 3rd " is also W

Q8. The odds against a certain event are 3 to 2 & in favour is independent of the former event are 4 to 3. Find the odds ^{for or} against their happening together [27 to 8]

Q9. 3 people A, B, C have P $\frac{4}{5}$, $\frac{2}{5}$ & $\frac{3}{4}$ of hitting a target. Find P that 3 win with exactly 2 hits if A, B, C each fires once at target. [$\frac{9}{20}$]

Q10. A bag has 4 R & 3 B. 2 drawings of 2 balls are made. Find P of drawing:

- i) 2 R balls
- ii) 2 B balls. [2/35]
- a) If ball is returned ~~after~~ after 1st draw
- b) not returned

Q11. In a ball factory, M₁, M₂, M₃ manufactures 25%, 35% & 40%. Of their output, 5%, 4% & 2% are defective. 1 is chosen &

Q11. From a set of n balls, 1, 2, 3, ..., 17, 1 ball is drawn at random. Find P that multiple of 3 or 7, [7/17]

Q12. 2 persons, A and B stand in a line with 10 other persons. Find P that there are 3 persons b/w them. [4/33]

Random Variables

A real variable associated with an outcome random exp. that is a var. takes a definite set of real values with definite set of prob. This is random var.

Eg. Assigning random var. from a set to random exp.

If a coin is tossed, coins obtained is either 0 or 1.

If n are taken at random ~~var~~ which is either good or bad (head or tail), $0, 1, 2, \dots, n-1$ (or n). We thus associate a var. with exp. that takes definite value, this is called random var.

Let X be random variable, $x_1, x_2, x_3, \dots, x_n$

with P $p_1, p_2, p_3, \dots, p_k$ i.e. $\sum_{i=1}^k p_i = 1$, so

$$\begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ n \end{bmatrix} \rightarrow \begin{pmatrix} T & T & T & \dots & n^{\text{th}} \\ H & T & T & \dots & n^{\text{th}} \\ \vdots & & & & \\ H & H & H & \dots & n^{\text{th}} \end{pmatrix}$$

Eg 1
If a pocket die is thrown, no. of coins obtained (X) & prob. distribution as

$$P_X(x) : \begin{matrix} 1 & 1 & 1 & \dots & 1 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \dots & \frac{1}{6} \end{matrix}$$

Eg 2
In tossing 2 coins $\{ (H, H), (H, T), (T, H), (T, T) \}$,

Miss

If 13 cards are chosen from a deck, the no. of aces among a random var., say X , it's possible values, $0, 1, 2, 3$ and 4 . So, prob. that $X=0$ $\left(\frac{48C_3}{52C_3} \right)$ & prob. that $X=1$ $\left(\frac{4C_1 \times 48C_2}{52C_3} \right) \dots$

A random var. which consists finite & or infinite no., a value is called discrete random var.

If a random var. consists infinite value, - -

• Population & sample

To answer any statistical knowledge about some aggregate of individual object, this is called population ~~with~~ which is as large whereas data in hand relate to only a small no. called sample, i.e. entire set of objects & data points is called population, any subset $\rightarrow$ sample

• Theoretical

It is a statistical object that represent prob distribution derive from theoretical rather empirical. It describes a random process & how freq it is to occur.

It predicts about the population, to get sample.

• Likelihood

$\rightarrow$ To get given set of data, statistical model, how likely set of parameter value produce the observed data. Thus, theoretical distribution defines the P of diff. outcomes on set & how certain parameters are.

Let P of coin landing head is 'p' which follows binomial. Now, 5 times $\rightarrow$ 3 H & 2 T.

Theoretical (EMF) = $\sum_{x=0}^n p^x (1-p)^{n-x}$ Observed outcome,

X = random var.

Outcomes of toss & of head & tail.

? Miss

Let $L_p(P)$, $3 \times 8 \times 2^T$, $p^2(1-p)^2$, a func. expresses how likely the parameter is given data, $3 \times 8 \times 2^T$. Likelihood is func. of p & it tells how likely p are based on data.

Several likelihood func asked on given distribution func.

Discrete

A random var. which assumes finite or ∞ .

Eg Sample S:

x	y
1	$\frac{4}{9}$
2	1
3	0
4	1
5	$\frac{4}{9}$
6	$\frac{4}{9}$

So, random var. x take 6 values & $y \rightarrow 0, 1, 4, 9$.
 $x \rightarrow$ Discrete random var. which assumes, $x_1, x_2, \dots, x_n$ with $P_1, P_2, P_3, \dots, P_n$ where $\sum p_i = 1$.

Set of values define discrete prob. distribution,

f_x denotes prob., $f_x = \text{prob. } P(X=x)$ where

$f(x)$ is prob. mass func. of discrete random function of X . $f_x(x) \geq 0$, $\sum f_x(x) = 1$, i.e.

$$f(x) = \begin{cases} p_i & \text{where } X=x_i \\ 0 & \end{cases}$$

Continuous

A random var. that assumes an uncountable inf. no. of values.

If X is cont. random var., no. of possible val. which is cont. is ∞ , where $f(x)$ called P.D.F. satisfy:

$$1. f(x) \geq 0$$

$$2. \int_{-\infty}^{\infty} f(x) dx = 1$$

1. The 'P' that ^{random var.} it lies b/w c and d , ($c \leq x \leq d$)

$$= \int_c^d f(x) dx$$

* The prob that random X takes values $\leq x$ is func. of x denoted $F(x)$ called C.D.F., ↓
Specified value

$$F(x) = \text{Prob.}(X \leq x) \text{ where } 0 \leq F(x) \leq 1$$

For discrete prob. distribution X , ^{assuming} ~~any~~ finite no. $x_1, x_2, \dots, x_n$ with prob. $p_1, p_2, \dots, p_n$ then C.D.F. is given by

$$F(x) = 0 \text{ when } x \leq x_1, \text{ then } F(x) = p_1 \text{ when } x_1 \leq x < x_2$$

$$\& F(x) = p_1 + p_2 \text{ when } x_2 \leq x < x_3$$

Similarly, so on...

$$F(x) = p_1 + p_2 + \dots + p_n = 1 \text{ when } x_n \leq x$$

8 $F(x) = p_1 + p_2$ when $x_2 \leq x < x_3$

Similarly, so on...

$$F(x) = p_1 + p_2 + \dots + p_n = 1 \text{ when } x_n \leq x$$

9/1/20

The prob. that X lies in a interval say x_1 and x_2 (inclusive both) is ~~$P(x_1) \leq X \leq x_2$~~ ~~$= P(X) \leq P(x \leq x_1)$~~

$$\text{C.D.F. } P(x \leq x_2) - P(x \leq x_1) = \text{C.D.F.}(x_2) - \text{C.D.F.}(x_1)$$

For a continuous prob. distribution with P.D.F. $f(x)$ in $a \leq x \leq b$, C.D.F. is given by $F(x) = \int_a^x f(z) dz = \text{P.D.F. of } f(x)$ can be obtained by C.D.F. $F(x) = \frac{d}{dx} (F(x))$.

If C.D.F. is known, the probabilities of the given intervals can be obtained as $p(c \leq x \leq d) = F(d) - F(c)$.

Joint distribution of 2 variables x and y describes prob. distribution that modules likelihood of 2 random var. occ. simult. It provides prob. that each combination of 2 var. will occur.

• For discrete, X & Y , joint is expressed as $P(X=x, Y=y)$ which gives prob. corresponds to the distribution, & co-var. dist. func. is $\rho(X=Y)(x, y)$, joint dist. allows understand how 2 var. are related or independent.

Q1) Is the following a P.D.F. where $f(x) =$

Basically:

$$f(x) = \begin{cases} 2x, & 0 < x \leq 1 \\ 4-2x, & 1 < x \leq 2 \\ 0, & \text{else} \end{cases}$$

$$\begin{cases} 2x, & 0 < x \leq 1 \\ 4-2x, & 1 < x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

b) If random var. X has P.D.F $f(x) = \begin{cases} \frac{1}{4}, & x \in [-2, 2] \\ 0, & \text{else} \end{cases}$ obtain $2x+3 > 5$.

Ans. a) If $f(x)$ is P.D.F., it must

satisfy, $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x) dx = 1$

$$= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^{\infty} f(x) dx$$

$$= 0 + \int_0^1 f(x) dx + \int_1^2 (4-2x) dx + 0$$

$$= 1 + 4 - 3 = 2$$

Ans b) $2x+3 > 5 \Rightarrow x > 1$

$$\text{So, } P(2x+3 > 5) = P(x > 1) = \int_1^2 f(x) dx + \int_2^{\infty} f(x) dx$$

$$= \int_1^2 \frac{1}{4} dx + 0 = \frac{1}{4}$$

Q2 An unbiased coin is thrown 3 times. If random var. X denotes no. of heads. Find C.D.F. of X .

Soln:

$x:$	0	1	2	3
$f(x):$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$F(0) = P(X \leq 0) = P(X=0) = \frac{1}{8}$$

$$F(1) = P(X \leq 1) = P(X=0) + P(X=1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$F(2) = P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$$

$$\therefore F(3) = P(X \leq 3)$$

Thus,

$x:$	0	1	2	3
$F(x):$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{7}{8}$	1

Q3. The P.D.F. of cont. random var. is $y = K(x-1)(2-x)$

for $1 \leq x \leq 2$. Determine:

- i) K ii) Find C.D.F. iii) Prob. that $X \leq \frac{3}{4}$ iv) Prob. where $X > \frac{3}{2}$
 v) Prob. when $\frac{3}{4} \leq X \leq \frac{3}{2}$

Soln: $y = K(3x - x^2 - 2)$

$$i) \int_1^2 f(x) dx = 1 \Rightarrow \int_1^2 K(3x - x^2 - 2) dx = 1$$

$$\therefore \boxed{K=6}$$

$$ii) F(x) = \int_1^x f(t) dt = 5 - 12x - 9x^2 - 2x^3 \text{ as } 1 \leq x \leq 2$$

$$iii) P(X \leq \frac{3}{4}) = F(\frac{3}{4})$$

$$iv) P(X > \frac{3}{2}) = 1 - P(X \leq \frac{3}{2}) = 1 - F(\frac{3}{2})$$

$$\begin{aligned}
 \therefore P\left(\frac{9}{4} \leq X \leq \frac{9}{2}\right) &= P\left(X < \frac{9}{2}\right) - P\left(X < \frac{9}{4}\right) \\
 &= F\left(\frac{9}{2}\right) - F\left(\frac{9}{4}\right) \\
 &= \frac{11}{32}
 \end{aligned}$$

* Expected variance & standard deviation

Let X be random var. with distribution func. f of X .

$$\text{So, } f(x_i) = P(X = x_i)$$

$$x: \quad x_1 \quad x_2 \quad x_3 \quad \dots$$

$$f(x): f(x_1) \quad f(x_2) \quad f(x_3) \quad \dots$$

So, mean/expectation/expected value of X is denoted by $E(X)$

is $E(X) = \sum_{i=1}^{\infty} x_i f(x_i)$ if X is discrete random var.

$\int_{-\infty}^{+\infty} x f(x) dx$ if X is ~~con~~ cont. random variable.

Let $\mu = E(x)$, so var. of $X = \text{Expectation of } (x - \mu)^2$

$$= \sum_i (x_i - \mu)^2 f(x_i) \text{ if } x \text{ is discrete.}$$

$$\int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx \text{ if } X \text{ is cont. random var.}$$

$$\text{Thus, standard deviation} = \sqrt{\sigma_x}$$

If x is discrete random var., then

$$(x_1 - \mu)^2 f(x_1) + (x_2 - \mu)^2 f(x_2) + \dots + (x_n - \mu)^2 f(x_n)$$

$$= \text{Expectation of } x^2 - 2(\mu \times \mu + \mu^2)$$

$$\sum_i f(x_i) = 1 = \text{Expectation of } x^2 - \mu^2$$

$$= \text{" of } x^2 - (\text{Expectation of } x)^2$$

Cor.

$$1. \text{ Var. of } a(x+b) = a^2 \text{ var.}(x)$$

$$\sigma(ax+b) = |a| \sigma(x).$$

2. Var. of $X + K = \text{Var. of } X$

σ of $X + K = \sigma$ of X

3. Var. of $KX = K^2 \text{Var. of } X$

$\sigma_{XY}(KX) = \text{if } X \text{ and } Y \text{ are independent, then}$
 $= |K| \sigma(X)$

Expectation of $X \times Y = \text{Expect. of } X \times \text{Expect. of } Y$.

Eg.
 Let X be random. var. with P.D.F. $f(x) = \begin{cases} \frac{1}{2}x, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$

Find expectation of X $[E(X)]$ & var. of X .

Soln: $E(X) = \int_{-\infty}^{\infty} xf(x)dx$

$= \int_0^2 \frac{1}{2}x^2 \cdot dx + 0 = \frac{8}{6} = \frac{4}{3}$

Expectation of random. var. $X^2 = \int_{-\infty}^{\infty} f(x)dx$

$= \int_0^2 \frac{1}{2}x^3 dx$

$\therefore \text{Var. of } X = \text{Expect. of } X^2 - (\text{Expect. of } X)^2$

$= 2 - \frac{16}{9} = \frac{2}{9}$

$\therefore \sigma = \frac{\sqrt{2}}{3}$

Eg. The P.D.F. of $f(x) = 0$ at 0, 1, 2 & $f(0) = c, f(1) = 2c - 3c^2$

& $f(2) = 4c - 1$. Find :

i) c ii) Cond. prob. $P(X \geq 1)$ when $X < 2$

iii) Expectation & variance of X .

Soln: i) $\sum f(x) = c + 2c - 3c^2 + 4c - 1 = 0$

$= (c-2)(3c-1) = 0$

$\therefore c = 2, \frac{1}{3}$

* $C=2$, $f(x) \geq 0$ & $\sum f(x) = 1$ are not satisfied as each $0 \leq f(x) \leq 1$ so $C = \frac{1}{3}$.

ii) $\frac{P(X \geq 1) \text{ and } X < 2}{P(X < 2)}$

$$= \frac{P(X=1)}{P(X=0) + P(X=1)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3}} = \frac{1}{2}$$

iii) Expectation of $X = 0 \times \frac{1}{3} + 1 \times \frac{1}{3} + 2 \times \frac{1}{3} = 1$

$$\text{Var. of } X = 0^2 \times \frac{1}{3} + 1^2 \times \frac{1}{3} + 2^2 \times \frac{1}{3} - 1^2 = \frac{5}{3} - 1 = \frac{2}{3}$$

H.W. (IMP)
Q1. A random var. X has P.D.F. as $f(x) = \begin{cases} cx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$

Find 'c' and $P(0 \leq X \leq \frac{1}{2})$, $P(X > \frac{3}{4})$

& $P(\frac{1}{4} < X < \frac{3}{4})$. Also obtain C.D.F.

$$[C = 3; \frac{1}{8}, \frac{37}{64}, \frac{13}{32}; F(x) = x^3] \text{ Ans.}$$

Q2. Show $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ K-x, & 1 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$

$$[x=2]$$

For what suitable

Q3. In ~~cont.~~ cont. var., $f(x) = \frac{1}{2} - kx$, $0 \leq x \leq 4$, where

a is const. Find i) a ii) Prob. that X lies $\in [\frac{2}{3}]$

iii) Find C.D.F. & calc. $P(X \leq 1)$, $P(X \geq 2.5)$, $P(|x-2| \leq 0.5)$

$$[\frac{1}{8}, \frac{3}{16}; \frac{x(8-x)}{16}, \frac{7}{16}, \frac{9}{64}, \frac{1}{64}]$$

Q4. A random. var. has follow. dist.

$x:$ 4 5 6 8

$P:$ 0.1 0.3 0.4 0.2

$$[3.9, 1.22]$$

Find expectation & standard var.

Q5. For what of a , $f(x) = \frac{a}{x^2}$, where $x = 1, 2, 3, \dots, n$ be the prob mass func of discrete random var. x . Find mean & var. of X .

Q6. A random var. X is defined:

$$P(X=1)=p \text{ and } P(X=0)=1-p.$$

Find mean & var.

Q7. Obtain exp. of no. of tails preceding first head in a indefinite series of tosses of some coin.

Q8. X is discrete random var. has prob. mass. func.

$x:$	0 0	1	2	3	4	5	6	7
$P(X=x):$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

i) Determine k

ii) Find $P(X < 6)$

iii) Val. of $P(X \geq 6)$?

$$[0.1 ; 0.81 ; 0.93]$$

Distribution [CO2]

→ 3 discrete, cont., poisson

Discrete (Uniform): The 2nd Prob. dist. of random var. (R.V. (X)) with n possible values $x_1, x_2, x_3, \dots, x_n$ with equal prob. is called uniform discrete given as:

~~$x_1, x_2, \dots, x_n$~~

$$x: x_1 \quad x_2 \quad x_3 \quad \dots \quad x_n$$

$$f(x): \frac{1}{n} \quad \frac{1}{n} \quad \frac{1}{n} \quad \dots \quad \frac{1}{n}$$

etc:

- ① P.M.F. of discrete is constant (independent of x).
- ② Possible value of R.V. may be any real no. & succ. values may not have a common diff.

1. If discrete r.v. X follows uniform dist. & assumes only values 8, 9, 11, 15, 18, 20. Find prob.:

$$\begin{aligned} \text{i)} P(X=9) \quad \text{ii)} P(X=12) \quad \text{iii)} P(X < 15) \quad \text{iv)} P(X \leq 15) \\ \text{v)} P(X > 15) \quad \text{vi)} P(|X-14| < 5) \end{aligned}$$

Ans: $\because X$ is R.V. with discrete of 6 possible value.

Prob. of each value $= \frac{1}{6}$ & prob. of other value (than the given) is 0.

$$\text{So, i)} P(X=9) = \frac{1}{6} \quad \text{ii)} P(X=12) = 0 \quad \text{iii)} P(X < 15) = \frac{1}{6} \times 3 = \frac{1}{2}$$

$$\text{iv)} P(X \leq 15) = \frac{1}{6} \times 4 = \frac{2}{3} \quad \text{v)} P(X > 15) = \frac{1}{6} \times 2 = \frac{1}{3}$$

$$\begin{aligned} \text{vi)} P(|X-14| < 5) &= P(X=15) + P(X=18) + P(X=20) \\ &= \frac{1}{6} \times 3 = \frac{1}{2} \end{aligned}$$

2. Find mean & S.D. of uniform dist. $f(x) = \frac{1}{n}$ where

$$n=x. \quad x=1, 2, 3, \dots, n$$

$$\text{So, mean} = (\mu) = \sum x f(x) \quad \text{for } x=1, 2, 3, \dots, n$$

$$\sum_x f(x) = \frac{1}{n} \sum_1^n x = \frac{n+1}{2}$$

↓
Expect. of X

$$\text{Expec of } X = \sum x^2 f(x) = \frac{1}{n} \sum x^2 = \frac{1}{n} \times \frac{n(n+1)(2n+1)}{6} \quad \text{Check}$$

$$= \frac{(n+1)(2n+1)}{6}$$

$$\text{Var. of } X = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{n^2-1}{12}$$

$$\therefore \text{S.D.} = \sqrt{\frac{n^2-1}{12}}$$

* Binomial distribution

A theoretical (discrete) which has n independent trials (such trials are Bernoulli's trial), each with 2 possible outcome, success & failure, i.e., sample size of n be drawn at random with replacement.

Let X = random var. indicate no. of success in such exp.

$$\begin{array}{ccccccc} x: & 0 & 1 & 2 & 3 & \dots & n \\ f(x): & q^n C_0 & q^{n-1} p C_1 & q^{n-2} p^2 C_2 & \dots & p^n C_n \end{array}$$

Thus, P.M.F is $f(x) = {}^n C_x p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$

So, mean = expectation of X ($p+q=1$)

$$= \sum_{i=0}^n x_i f(x_i)$$

$$\begin{aligned} &= 1 \times {}^n C_0 q^n + 2 \times {}^n C_1 q^{n-1} p + \dots + n \times {}^n C_{n-1} q p^{n-1} \\ &= n p (q^{n-1} + (n-1) C_1 + \dots + n-1 C_{n-1} q^{n-1}) \\ &= n-1 C_{n-1} q^{n-1} \\ &= n p \quad (p+q=1) \end{aligned}$$

Check ?

For Bernoulli's r.v. Y ,

Expec. of $Y = p$

Var. of $Y = pq$, where $Y = \begin{cases} 1, & \text{Success} \\ 0 & \text{failure} \end{cases}$

• Expec. of $Y = 1 \times P(Y=1) + 0 \times P(Y=0)$
 $= 1 \times p + 0 \times q$
 $= p$

• Var. of $Y = E(Y^2) - (E(Y))^2$, $Y^2 = \begin{cases} 1^2=1, & \text{if } Y=1 \\ 0^2=0, & \text{if } Y=0 \end{cases}$
 $= p(Y^2) - (E(Y))^2$
 $= p - p^2$
 $= p(1-p) = pq$

* Var. of X :

Let $X = X_1 + X_2 + X_3 + \dots + X_n$, $X_i = \text{Bernoulli's r.v.}$

$X = 0, 1$ for failure or success

Expec. of $X_i = p$

Var. of $X_i = pq$

$\therefore$ Var. of $X = (\text{Var. of } X_1) + \text{Var.}(X_2) + \dots + \text{Var.}(X_n)$
 $= \text{Var.}(X_1) + \text{Var.}(X_2) + \dots + \text{Var.}(X_n)$
 $= npq$ $X_i = pq$
(All i)

Also,

Var. of $X = E(X^2) - (E(X))^2$

$\textcircled{*} = E(X^2) \cdot f(x) \leq x^2 \leq x(x-1) + x$

$= \sum x(x-1)f(x) + E(X)$

$= \sum_{x=0}^n x(x-1)f(x)$

$= 2x1x^n C_2 p^2 + p^2 \times 2x1 \dots$

$= 2x1x^n C_2 p^2 + 3x2x^n C_3 + pq \cdot n$
 $+ 4x3x^n C_4 p^4 q^{n-4}$

$\checkmark = n(n-1)p^2 [q^{n-2} + q^{n-3} \cdot p \cdot (n-2)q^{n-4} \cdot p \cdot \frac{(n-2)}{(n-2-1)} + \dots p^{n-2}]$

$$= n(n-1)p^2(q+p)^{n-2}$$

$$= n(n-1)p^2 \quad (\because p+q=1)$$

$$\begin{aligned} \text{Var. of } X &= E(X^2) - (E(X))^2 \\ &= E(X(X-1)) + E(X) - (E(X))^2 \\ &= n(n-1)p^2 + np - n^2p^2 \\ &= npq \end{aligned}$$

1. 5 coins are tossed 3200 times. Find exp. freq of ~~prob~~^{dist.} of heads & tails & tabulate result. Calc. mean no. of heads & tails

Soln: $\therefore$ Coins is unbiased.

$$p = \frac{1}{2} (\text{head}) \quad \& \quad q = \frac{1}{2} (\text{tail})$$

h	t	Prob.	Exp. freq.
0	5	$1/32$	100
1	4	$5/32$	500
2	3	$10/32$	1000
3	2	$10/32$	1000
4	1	$5/32$	500
5	0	$1/32$	100
		1	3200

$$\therefore \text{Mean} = \text{Exp. of } X = 5 \times \frac{1}{2} = 2.5$$

$$\text{S.D.} = \sqrt{\text{Var.}(X)} = \sqrt{5 \times \frac{1}{2} \times \frac{1}{2}} = \frac{\sqrt{5}}{2}$$

No. of heads	Freq.
0	100
1	500
2	1000
3	1000
4	500
5	100

Q2. In 10 independent throw of def dice, prob. that even no. will appear 5 times is twice the prob. that even no. will appear 4 times. Find P that even no. will NOT appear.

Soln: Let p = Prob. of getting even no.
 q = " of " odd no.

P.M.F. . $f_x(x) = {}^{10}C_x p^x q^{n-x}$, where x = Appearing even no.

$$f(5) = 2 \times f(4)$$

$$\Rightarrow {}^{10}C_5 p^5 q^5 = 2 \times {}^{10}C_4 p^4 q^6$$

$$\therefore \boxed{q=3} \quad 3p = 5q$$

$$\Rightarrow 5(1-p) = 3p \quad \therefore p = \frac{5}{8}, q = \frac{3}{8}$$

$$\therefore f(0) = {}^{10}C_0 p^0 q^{10-0} = \left(\frac{3}{8}\right)^{10}$$

Q3. The overall % of failures in a certain exam
 What is prob. that out of 6, atleast 4 pass the exam?
 Soln: Let p be prob. of success in a single trial is p
 & prob. of failure $= 40\% = \frac{2}{5}$

$$= p = 1 - \frac{2}{5} = \frac{3}{5}$$

$n = \text{No. of candidates} = 6$

$$\therefore \text{Prob. of } x \text{ success} = f(x) = {}^nC_x p^x q^{n-x}$$

$$= {}^6C_x \left(\frac{3}{5}\right)^x \left(\frac{2}{5}\right)^{6-x}$$

$\therefore \text{Prob. atleast 4 pass}$

$$= f(4) + f(5) + f(6)$$

$$= {}^6C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^2 + {}^6C_5 \dots + {}^6C_6 \dots$$

$$= \frac{1701}{3125}$$

* Standard discrete distribution

A discrete dist. with cmf defined as

$$f_x(x) = \frac{e^{-\lambda} \lambda^x}{x!} \cdot 0, \quad x = 1, 2, 3, \dots, \infty$$

$\mu_0 \rightarrow \text{Confirm once}$

$\lambda = \text{ave no. known as parameter of dist.}$

Sometimes, referred as rate parameter

$\Rightarrow$ Avg. no. of events occurring in fixed time dist

$\therefore$ Poisson's dist. is prob. of no. of events happening in that interval s.t. event occurs indep. with avg weight λ .

Poisson's ratio:

$x:$	0	1	2	3	...	∞
			$\downarrow$	$\downarrow$		
$f_x(x)$	$e^{-\lambda}$	$e^{-\lambda} \cdot \lambda$	$\frac{e^{-\lambda} \cdot \lambda^2}{2!}$	$\frac{e^{-\lambda} \cdot \lambda^3}{3!}$		

$$\sum f_x(x) = 1$$

Properties:

① R.V. corresponds to dist. assumes ∞ no. of values $0, 1, 2, \dots, \infty$.
Dist. is completely specified when λ is known.

② $\text{Mean} = \text{Var.} = \lambda$

③ If x & y are independent poisson's variables with λ_1 & λ_2 respect., then X & Y also follows the same

④ Poisson's dist. may be used as approx. to binomial dist. when $p \rightarrow \text{small}$, $n \rightarrow \text{large}$ but $np \rightarrow \text{finite}$.

~~Ex~~ $\sum_{x=0}^{\infty} f(x, \lambda) = 1$

$$\Rightarrow \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$\Rightarrow e^{-\lambda} \left(\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots \right)$$

Poisson's vs Binomial

Poisson's dist. may be obtained as a limiting case of bin.:

• $n \rightarrow \infty$, prob. of success $\rightarrow 0$ (tends), Mean is finite

$$\text{So, } {}^nC_x \cdot p^x \cdot q^{n-x} \approx \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

Prove:

$${}^nC_x p^x q^{n-x} = n(n-1)(n-2)\dots(n-x+1) \cdot p^x \cdot q^{n-x}$$

$$= np^x \frac{1 \cdot (1 - \frac{1}{n})(1 - \frac{2}{n}) \dots (1 - \frac{x-1}{n})}{x!}$$

$$= \left(1 - \frac{np}{n}\right)^{n-x}$$

$$np = \lambda$$

Plz check

?

$$\text{If } x \rightarrow \lambda, \frac{1}{x} \rightarrow 0$$

$$\Rightarrow \text{Exp.} = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \rightarrow 1$$

$$\text{So, } {}^nC_x \cdot p^x \cdot q^{n-x} = \frac{\lambda^x}{x!} \left[\left(1 - \frac{\lambda}{n}\right)^{n-x} \right]$$

$$= \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$\text{If } n \rightarrow \lambda, \frac{1}{\lambda/n} \rightarrow 1 \quad \& \quad \left(1 - \frac{\lambda}{n}\right)^{n-x} \rightarrow \frac{1}{e}$$

$$\text{So, } {}^nC_x \cdot p^x \cdot q^{n-x} = \frac{\lambda^x \cdot e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

*Find mean & var. of poisson's disto is given,

$$\text{Let } f_x(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

λ = Real parameter,

$$\text{Mean} = E(x) = \sum_{x=0}^{\infty} x f_x(x)$$

$$\sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} = 0 \cdot f(0) + 1f(1) + 2f(2) + \dots \infty$$

$$\Rightarrow 0 + 1 \cdot \frac{e^{-\lambda} \cdot \lambda}{1!} + 2 \cdot \frac{e^{-\lambda} \cdot \lambda^2}{2!} + \dots$$

$$= \lambda \cdot e^{-\lambda} \left(1 + \frac{\lambda}{2} + \frac{\lambda^2}{2 \cdot 2} + \dots \right)$$

$$E(x \cdot (x-1)) = \sum_{x=0}^{\infty} x(x-1) \cdot f_x(x)$$

$$= 0f(0) + 1 \times 0 \times f(1) + 2 \times 1 \times f(2) + 3 \times 2 \times f(3) + \dots \infty$$

$$= \frac{2 \cdot e^{-\lambda} \cdot \lambda^2}{2!} + \frac{3 \cdot 2 \cdot e^{-\lambda} \cdot \lambda^3}{3!} + \frac{4 \times 3 \cdot e^{-\lambda} \cdot \lambda^4}{4!} + \dots \infty$$

$$= e^{-\lambda} \cdot \lambda^2 \cdot \left[1 + \frac{\lambda}{2!} + \frac{\lambda^2}{3!} + \dots \infty \right]$$

$$= \lambda^2$$

$$\begin{aligned}
 \therefore \text{Var.}(x) &= \text{Exp}(x^2) - (E(x))^2 \\
 &= E(x(x-1)) + E(x) - (E(x))^2 \\
 &= \lambda^2 + \lambda - \lambda^2 \\
 &= \lambda
 \end{aligned}$$

Poisson's process: Math. model that describes series of events occ. randomly over time:

- ① Events occur independently
- ② Prob. of event happening given time interval depends on length of interval, not where it starts.
- ③ Events happen once $\Rightarrow$ 2 events can't occur at same time
- ④ Event occurs at avg. rate called intensity or rate parameter as λ which is exp. no. of events per unit time

* It is charac. by events by $N(t)$ in ~~closed~~ time interval $[0, t]$ where $N(t)$ follows Poisson's.

Prob. of obser. K events in time t is given by:

$$P(x) = N(t) = K = \frac{(\lambda t)^K \cdot e^{-\lambda t}}{K!}$$

Thus, mean of $L(t) = \lambda \cdot t$
 Var. of $L(t) = \lambda \cdot t$

Q1. A r.v. X follows a Poisson's d.s. with parameter 3.

Find prob. that variable ass. value:

- i) 0, 1, 2 and 3 ii) Less than 3 iii) Atleast 2

Soln: Let $f_x(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$$\frac{e^{-3} \cdot 3^x}{x!}$$

$$f(0) = 0.0498$$

$$i) f(1) = \frac{3^1 \cdot e^{-3}}{1!} \quad f(2) = \frac{e^{-3} \cdot 3^2}{2!} \quad f(3) = \frac{e^{-3} \cdot 3^3}{3!}$$

$$\begin{aligned}
 ii) f(0) + f(1) + f(2) &= 0.0498 + 0.1494 + 0.2241 \\
 &= 0.4433
 \end{aligned}$$

$$\text{iii) } f(0) + f(1) + \dots = 1$$

$$\Rightarrow f(2) + f(3) + \dots = 1 - (f(0) + f(1))$$

$$= 1 - 0.1192$$

$$= 0.8808$$

Q2. A business firm in a subject receives in an avg. 2.5 telephone calls per day during 10.00 to 10.05 a.m. Find P on certain day, firm receives:

i) No calls ii) Exactly 4 calls (During same period)

Ans. Let X = no. of calls recd. 10.00 to 10.05 a.m. which follows dist. $\lambda = 2.5$

$$f_x(x) = \frac{e^{-2.5} \cdot (2.5)^x}{x!}$$

$$f_x(0) = 0.0821$$

$$f_x(4) = 0.1336$$

3) Let X be distributed in the poisson form. If $P(X=1) = P(X=2)$

What is $P(X=1)$ or 2 ? find exp of X

ans $f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ given $\frac{e^{-\lambda} \lambda}{1} = \frac{e^{-\lambda} \lambda^2}{2!}$ as $P(X=1) = P(X=2)$

$$\Rightarrow \boxed{\lambda = 2}$$

So $P(X=0 \text{ or } 1) = P(X=0) + P(X=1)$

$$= \frac{e^{-\lambda} \lambda^0}{0!} + \frac{e^{-\lambda} \lambda^1}{1!} = 3e^{-2}$$

exp $= \lambda = 2$

4) Two per cent of the item are def. Find prob 3 or more are def in a sample of 100 items

The no. of def (X) follows binomial distr since $p = 2\% = 0.02$

$n = 100$, is fairly large, making $np = 2$ a finite quantity

we use poisson distribution $\lambda = 2$

$$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

So probability (3 or more) = $f_X(3) + f_X(4) + f_X(5) + \dots$

$$= 1 - \{f_X(0) + f_X(1) + f_X(2)\}$$

$$= 1 - \left(\frac{e^{-2} 2^0}{0!} + \frac{e^{-2} 2^1}{1!} + \frac{e^{-2} 2^2}{2!} \right)$$

$$= 0.325$$

Poisson Process

A server processes requests from customers.

The no of requests received by the server follows a poisson process with a rate of 6 ($\lambda = 6$) requests per hour. Each request is handled independently and the service time are exponentially distributed with an avg of 10 minutes per request.

1) $P(\text{receiving exactly 5 requests in hour})$ 2) $P(\text{that the time until the next request after 1 hour will be more 15 minutes})$

1) ~~P(exactly 5)~~ $\Rightarrow P(\text{that the time } 3) \text{ If we start observing at 3 p.m.}$
 what is the prob that more than 12 req will be received b/w 3 p.m to 5 p.m.

Ans $\lambda = 6$ so, $\lambda_{2 \text{ hour}} = 6 \times 2 = 12$

$$P_{mf} = P(X=K) = \frac{\lambda_{2 \text{ hour}}^K e^{-\lambda_{2 \text{ hour}}}}{K!}$$

$$P(X=5) = \frac{12^5 e^{-12}}{5!} = 0.093$$

2) The time b/w requests in a Poisson process follows an expo distribution with rate $\lambda = 6$ per hour.

$$15 \text{ min} = 0.25 \text{ h}$$

$$P(\text{what } t > 0.25) = e^{-\lambda t} = e^{-6 \times 0.25} = e^{-1.25} = 0.223$$

$$3) P(X \leq 12) = \sum_{k=0}^{12} \frac{\lambda_{2 \text{ hour}}^k e^{-\lambda_{2 \text{ hour}}}}{k!} = \sum_{k=0}^{12} \frac{12^k e^{-12}}{k!} = 0.715$$

$$P(X > 12) = 1 - P(X \leq 12) = 1 - 0.715 = 0.285$$

So there is 28.5% chance that more than 12 requests will be received b/w 3 to 5 p.m.

HW

1) A r.v. (X) follows P.D. with rate param $= 2$. Find the prob when $P(X=1)$, $P(X \leq 1)$, $P(X < 1)$, $P(X > 1)$, $P(1 \leq X \leq 3)$

2) A discrete r.v. (X) follows P.D such that $P(X=1) = P(X=2)$

Find mean and var of the distribution

3) A Poisson variant ~~Poisson~~ X takes +ve val is $1 - e^{-2}$ ~~$1 - e^{-2}$~~

Find mean, var, ~~$P(1 \leq X < 1.5)$~~

4) 3% bolts def, ~~P~~ $P(\text{sample of } 200 \text{ bolts will def})?$

5) on operator receives from no. of telephone calls 11:00 AM - 11:05 AM with $\lambda = 3$ (rate param)

1) $P(\text{no calls tomorrow})$ 2) $P(\text{next 3 days total of 1 call in that time interval})$

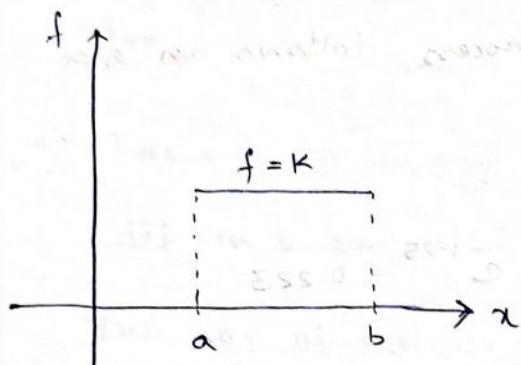
6) A car ~~has~~ ^{here} from ^{has} 2 cars. which it hires out day by day. The no of demands for each car on each day follows a

distribution with mean 1.5. Calculate the proportion of days on which
 1) Neither car is used 2) Some demand is refused.

Cont^s Distribution

Cont^s Uniform Dis Let X be a cont^s P.V follows a rand.

uniform distribution ~~in~~ in closed interval $[a, b]$, where $a < b$
 so the pdf is $f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$



$$k(b-a) = 1$$

$$\Rightarrow k = \frac{1}{b-a} = f_X(x)$$

$$\text{cdf } F_X(x) = \begin{cases} 0 & \text{when } x < a \\ \frac{x-a}{b-a} & \text{when } a \leq x \leq b \\ 1 & \text{when } x > b \end{cases}$$

Proof: $f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$

for $x < a$, $F_X(x) = 0$

for $a \leq x \leq b$, $F_X(x) = \int_a^x f_X(x) dx = \int_a^x \frac{1}{b-a} dx =$

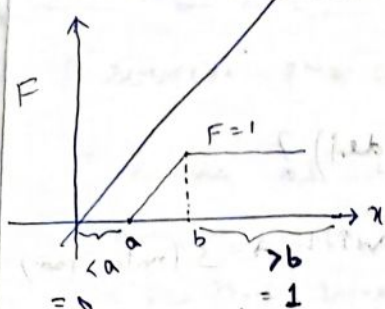
Now for $x < a$, $F_X(x)$ is 0, since the prob is 0

for $a \leq x \leq b$, $F_X(x) = \int_a^x f_X(x) dx = \int_a^x \frac{1}{b-a} dx = \frac{x-a}{b-a}$

for $x > b$, $F_X(x) = 1$

$$E(X) = \frac{a+b}{2}, \text{Var}(X) = \frac{(b-a)^2}{12}$$

Exponential Dis:



Interval time: It refers to the time elapsed b/w 2.

eg customers arrivals at store, phone

Proof

$$E(x) = \int_a^b x \cdot \frac{1}{b-a} dx = \left[\frac{x^2}{2(b-a)} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

Let x be the r.v. follows the,

~~pdf~~ $Var(x) = E(x^2) - [E(x)]^2$

$$E(x^2) = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \left[\frac{x^3}{3(b-a)} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 - ab + a^2}{3}$$

$$E(Var(x)) = \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2} \right)^2 = \frac{4b^3 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} \\ = \frac{a^2 - 2ab + b^2}{12} = \frac{(a-b)^2}{12} = \frac{(b-a)^2}{12}$$

Exponential Distribution

When exponential distribution is used?

Proof: ex/Inter-arrival time:

Let t be

It refers to the time lapse b/w the two consecutive events in a stochastic or random process/system. It is the time b/w the arrival one event and the arrival of next event

eg: customer arrivals at a store, phone calls at a call centre, or machine failure.

So, inter-arrival times are frequently associated with the Poisson process, which models the occurrence of events over time.

In this process, the no. of events occurring in a fixed time interval follows Poisson Distribution, that means inter-arrival times follows an exponential distribution.

Thus its rate param. of Poisson Process is λ .

The avg inter-arrival time = $\frac{1}{\lambda}$

Let X (R.V) follows the exponential dist with rate param λ

$$\text{Then pdf } f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \text{i.e. } X \sim \text{Exp}(\lambda)$$

Cdf of exponential distribution $F_X(x) = P(X \leq x)$

$$= \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$$

Thus $F_X(x)^{-1} = -\frac{\ln(1-x)}{\lambda}$ = inverse fun of cdf of the probability distr

Proof: let u be a rand var uniformly distn b/w 0-1
 i.e; $U \sim \text{Unif}(0, 1)$.

We need to find x s.t $f_X(x) = u$

$$\text{So; } U = f_X(x) = 1 - e^{-\lambda x} \Rightarrow x = -\frac{1}{\lambda} \log_e(1-u) \quad (??)$$

$$\text{thus } F_X^{-1} = -\frac{\log_e(1-u)}{\lambda} \quad (??)$$

Proof:

$$E(X) = -\lambda$$

$$\text{Var}(X) = \lambda^2$$

$$\text{then } E(X) = \int_0^{\infty} x \cdot \lambda \cdot e^{-\lambda x} dx \quad (\text{By parts})$$

$$= -x \cdot e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$= 0 + -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty}$$

$$+ E(X^2) = \int_0^{\infty} x^2 \cdot \lambda \cdot e^{-\lambda x} dx$$

$$= -x^2 \cdot e^{-\lambda x} \Big|_0^{\infty} + 2 \cdot \int_0^{\infty} x \cdot e^{-\lambda x} dx. \quad (\text{By parts})$$

$$= 0 + 2 \cdot \frac{1}{\lambda^2} = \frac{2}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Normal Distribution

16/1/26

Let r.v. X follows dist. called normal or Gaussian dist. defined as $X = (\mu, \sigma^2)$, $\mu \rightarrow$ Mean, $\sigma \rightarrow$ Variance. Then, P.D.F. is

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Standard normal dist.

Let X follows $N = \mu, \sigma^2$, the standard r.v. corresponds to X is defined by $Z = \frac{x - \mu}{\sigma}$ where $Z \rightarrow N(0, 1)$ when X lies in $(-\infty, \infty) \Rightarrow -\infty < Z < \infty$. So, $f_X(\sigma Z + \mu) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{Z^2}{2}}$
 $= f_Z(Z)$

Diff. $Z = \frac{x - \mu}{\sigma}$, we get $\sigma dZ = dx$

$$\text{So, } f_Z(Z) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{Z^2}{2}} \times \sigma = \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}} \quad ; e^{-\frac{Z^2}{2}} \text{ in}$$

~~$f_X(x)$~~ $f_X(x)$ should need the change to $\sigma e^{-\frac{Z^2}{2}}$

$$\phi(z) = f(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt$$

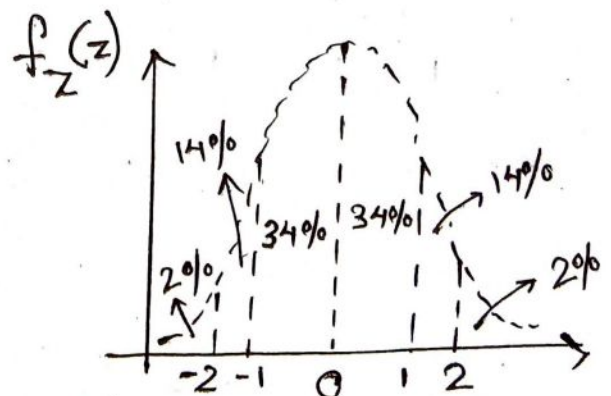
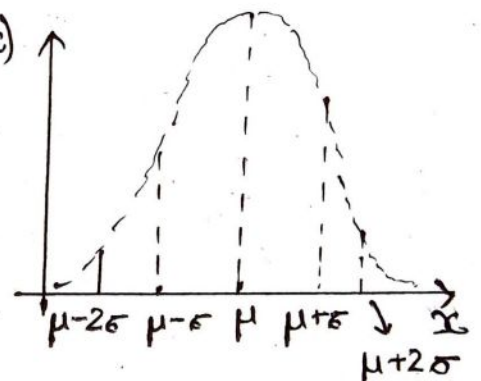
$\Rightarrow$ Prob. when $Z = z$

So, prob. when $A \leq Z \leq B$

$$= f(B) - f(A) = \phi(B) - \phi(A)$$

The identity rule: $f(-z) = \frac{1}{1 - f(z)}$

$$\text{or, } \phi(-z) = \frac{1}{1 - \phi(z)}$$



Note: $\therefore$ Normal prob. dist. is a closed form b/w any pairs of limit A & B , prob. relating to normal dist. is obtained from special tables such as standard normal tables.
 $\rightarrow$ Pertains to $\mu = 0$ & $\sigma = 1$ & its entries are values

of ^(check) $Z \sim e^{-z^2/2}$ $\Rightarrow$ Prob. when $Z=z \Rightarrow \phi(z)$
 we also used $f(-z) = \frac{1}{1-f(z)}$ which holds for all
 symm. dist. centered around 0. The cumulative prob
 $f(z)$ corresponds to area under standard normal density
 to the left.

Importance of normal dist:

The normal dist. has many imp. algebraic properties
 for which it is widely used in statistical theories. In
 many cases, physical/biological cases, data $\rightarrow$ normal dist.
 Statistical method in manufac. industry & theory of errors
 in measurements $\rightarrow$ normal dist. Also used to serve
 as approx. to binomial & tvc dist. In theory
 of sampling, widely used. Result. considerably simplifies
 work of testing & also useful for limits of parameters.

Approx. of binomial ^{with} normal

Binomial $B(n, p)$ can be approx. by normal $N(\mu, \sigma^2)$
 where $\mu = np$ & $\sigma^2 = npq$, p or q is not
 very small & n is sufficiently large ($np \geq 10$)
 or rather ($nq \geq 10$) which ensures $B(n, p)$ is not too
 skewed & can be approx. when $X \sim N(\mu, \sigma^2)$.

$\therefore$ Binomial dist $B(n, p)$ is discrete & $N(\mu, \sigma^2)$ is
 cont., we use continuity correction when approx. prob.
 i.e. binomial, $X = k$ is approx. in interval $[k-0.5, k+0.5]$

So, $P(K-0.5 \leq X \leq K+0.5) \approx K-0.5-\frac{\mu}{\sigma} \leq Z \leq K+0.5-\frac{\mu}{\sigma}$
 where Z is standard normal r.v. when parameter λ
 is large, poisson's $\approx$ Normal with same mean & S.D.
 as λ and $\sqrt{\lambda}$ respec. ($\lambda \geq 10$).

Q1. Binomial approx. using normal $N=100$ (No. of trials),
 $p=0.4$ (success). Approx. of getting 40 success $P(40)$.
Soln $np=0.4 \times 100=40$, $nq=60$

So, $np > 10$ & $n(n-p) > 10$

So, normal dist.

Mean $= np = 40$, $\sigma = \text{S.D.} = \sqrt{np(n-p)} = \sqrt{24} \approx 4.9$

So, $B(100, 0.4) \approx$ with $N(40, 4.9^2)$

So, $P(39.5 \leq Y \leq 40.5)$, where Y is normal r.v.

To standardised, we use stand. normal when (Mean $= 0$
 & S.D. $= 1$) where $Z = \frac{X-\mu}{\sigma}$, So $Z(1) = \frac{39.5-40}{4.9}$

$\Rightarrow Z(1) \approx -0.102$ & $Z = +0.102$ So prob.

when $P(39.5 \leq Y \leq 40.5) = P(Z(2)) - P(Z(1))$

as $P(Z \leq 0.102) = \phi(0.102) = 0.541$ [will be provided]

$P(Z \leq -0.102)$
 $= \phi(-0.102) = 1 - \phi(0.102) = 1 - 0.541 = 0.459$

So, $P(39.5 \leq Y \leq 40.5) = 0.459$

Q2. The height dist. of group with 10000 men with height
 64.5 inch & S.D. $= 4.5$, Find no. of men whose height is:
 i) < 69 but > 59.5 ii) < 55.5 iii) > 73.5 inch

Soln: $\mu = 64.5$ inch, $\sigma = 4.5$ inch

i) % of men whose $55.5 < \text{height} < 69 =$

Area under standard normal with vertical lines
& corresp. to $\frac{55.5 - 64.5}{4.5} = -2$

$$Z \frac{69 - 64.5}{4.5} = 1, \text{ which is } 14\% + 34\% + 34\% \\ = 82\% \text{ of } 10,000 \\ = 8200$$

ii) % of men whose height < 55.5

= Area under standard normal to left of
value $= \frac{55.5 - 64.5}{4.5} = -2$ which is 2% of 10,000
 $= 200$

iii) % of men > 73.5

is area under to right $\frac{73.5 - 64.5}{4.5} = 2$

is 2% of 10,000 = 200

Q3. Mean height of 500 males is ¹⁵¹ pounds & S.D. is ^{1.5} pounds. Find no. of student with:

i) B/w 120 & 155 ii) More than 155 pounds where
wt. are normally dist. Given $\phi(0.27) = 0.6064$
& $\phi(2.7) = 0.9808$.

Soln: $\mu = 151$ & $\sigma = 1.5$

So, prop. of students b/w 120 & 155 = Area
under stand. normal curve b/w the vertical lines
& stand values $Z(120, 150) = -Z(1)$

$$Z(2) = \frac{155 - 150}{1.5} =$$

$$\text{So, } P(120 \leq X \leq 155) = P(-2.07 \leq Z \leq 0.27) \\ = \phi(0.27) - \phi(-2.07)$$

$$= 1 - \phi(2.07)$$

$$= 0.5872 \text{ [Check plz]}$$

ii) Prop. $x > 155$, where $x > 155$

$$= 1 - P(x \leq 155)$$

$$= 1 - P(Z \leq 0.27)$$

$$= 1 - \phi(0.27)$$

$$= 1 - 0.6064$$

$$= 0.3936$$

Q4. ~~Mean~~ Mean of normal dist. is 50 & 5% of values are > 60 .

Find, S.D. given area under stand. normal b/w $Z=0$ to

$Z=1.64$ is 0.43

Ans: The stand. normal curve = $\frac{10}{\sigma}$ $\xrightarrow{5\%}$ $\frac{60-50}{\sigma}$

Area to right of 0 =

Area b/w 0 & 1.64 = 0.43

Area to right of 1.64 = 0.05

$$\frac{10}{\sigma} = 1.64 \Rightarrow \sigma = 6.1$$

The lecture moved from a solved normal-approximation question (for a Poisson/Euler distribution) to the Central Limit Theorem, then to Chebyshev/Markov inequalities, then to a few continuous distributions (Log-normal, Gamma, Weibull), and finally homework questions.

Question (Poisson/Euler $\rightarrow$ Normal approximation)

Question statement:

A random variable X follows "Euler distribution" with mean 16
(i.e., $X \sim \text{Poisson}(\mu = 16)$).

Find $P(X \geq 20)$, given $f(0.875) = 0.8092$, where $f(\cdot)$ denotes the area under the standard normal distribution function.

Solution (step-by-step, as discussed):

1. Identify parameters for Poisson:

$\mu = 16$ and $\sigma = \sqrt{\mu} = \sqrt{16} = 4$.

(This is used because $\text{Poisson}(\mu)$ is approximated by Normal with mean μ and variance μ when μ is reasonably large.)

2. Apply continuity correction:

$P(X \geq 20) \text{ (Poisson)} = P(Y \geq 19.5)$

where $Y \sim N(\mu = 16, \sigma = 4)$.

(This is used because a discrete distribution is being approximated by a continuous normal distribution.)

3. Standardize using $Z = (x - \mu) / \sigma$:

$Z = (19.5 - 16) / 4 = 0.875$.

So $P(Y \geq 19.5) = P(Z \geq 0.875)$.

4. Convert right-tail probability using the given standard normal area:

$P(Z \geq 0.875) = 1 - \Phi(0.875)$

$= 1 - f(0.875)$

$= 1 - 0.8092$

$= 0.1908 = 0.19$.

Answer:

$P(X \geq 20) = 0.19$.

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a sequence of independent random variables with the same distribution, with mean μ and variance σ^2 .

Define the sample mean:

$\bar{X}_n = (1/n) \sum X_i$.

Then:

$Z_n = (\bar{X}_n - \mu) / (\sigma / \sqrt{n})$

approaches the standard normal distribution as n becomes large.

Hence:

$P(a \leq Z_n \leq b) \approx P(a \leq Z \leq b)$

where $Z \sim N(0, 1)$.

Inequalities (when distribution is unknown)

Chebyshev's and Markov's inequalities were introduced as tools to understand the behavior of random variables when an exact distribution (PMF/PDF) is not available.

They provide generally loose upper bounds on probabilities of deviation.

Chebyshev's Inequality

Let X be a random variable with mean μ and standard deviation σ .
For any $k > 0$,

$$P(\mu - k\sigma \leq X \leq \mu + k\sigma) \geq 1 - 1/k^2.$$

Equivalently,

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2.$$

Markov's Inequality

Let X be a non-negative random variable and let $a > 0$.
Then,

$$P(X \geq a) \leq E[X] / a.$$

It was stated that the upper bound from Markov's inequality is weaker than Chebyshev's inequality.

Some Continuous Distributions (PDFs)

Log-normal Distribution:

If $\ln(X) \sim N(\mu, \sigma^2)$, then X is log-normally distributed.

PDF:

$$f_X(x) = \frac{1}{x \sigma \sqrt{2\pi}} \cdot \exp\left\{-\frac{1}{2} \left[\frac{(\ln x - \mu)}{\sigma} \right]^2\right\}, x > 0$$

0, otherwise.

Gamma Distribution:

The Gamma distribution with parameters α, β has PDF:

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \alpha > 0, \beta > 0, x > 0$$

0, otherwise.

Here α is the shape parameter and β is the scale parameter.

Gamma function:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx.$$

Weibull Distribution:

The Weibull distribution has PDF:

$$f_X(x) = \alpha \beta x^{\beta-1} e^{-(\alpha x^\beta)}, \alpha > 0, \beta > 0, x > 0$$

0, otherwise.

α and β are shape parameters.

Homework (No solutions)

1. A die is tossed 1200 times. Find the probability that the number of sixes lies between 190 and 210, given:

Area between $z = 0$ and $z = 0.78$ is 0.2823,

Area between $z = 0$ and $z = 0.81$ is 0.2910.

2. In a sample of 120 workers in a factory, the mean and standard deviation of wages were Rs. 11.39 and 31.03 respectively.

Find the percentage of workers getting wages between Rs. 9 and Rs. 17 in the whole factory, assuming wages are normally distributed.

Given:

Area from $z = 0$ to $z = 0.78$ is 0.2823,

Area from $z = 0$ to $z = 1.86$ is 0.4686.

3. 5000 candidates appeared in an examination.

Minimum pass marks = 40.

Minimum distinction marks = 50.

Average marks = 43, SD = 7.

Find how many candidates are expected to get:

(a) simply pass

(b) distinction

Given:

$\Phi(0.43) = 0.6664$,

$\Phi(1) = 0.8413$.

4. Show that the probability that the number of heads in 400 tosses of a fair coin lies between 180 and 220 is approximately $2\Phi(2) - 1$.

5. The number of deaths per day due to road accidents and due to other causes independently follow Poisson distributions with parameters 2 and 6 respectively.

Find the probability that the total number of deaths on a particular day is 2 or fewer, given $e^{-4} = 0.18$.